

Rama Krishna Molakaseema Data Scientist

rama3017krishna@gmail.com | +1 (904) 907-0270 | USA | LinkedIn | [GitHub](#)

Summary

Results-driven Data Scientist with 3+ years of expertise in AI, ML, and data analytics, specializing in financial and healthcare domains. Proficient in developing predictive models, reinforcement learning strategies, and deep learning architectures to optimize decision making. Skilled in Python, SQL, and cloud-based deployments (AWS, GCP). Experienced in building ETL pipelines, real-time data processing, and interactive dashboards. Adept at stakeholder engagement, requirement gathering, and deploying scalable AI/ML solutions to drive business impact.

Technical Skills

- **Programming & Scripting:** Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch), SQL
- **Machine Learning & AI:** LSTM, Reinforcement Learning (DQN), SVM, Random Forest, XGBoost, Gradient Boosting, A/B Testing, NLP
- **Big Data & ETL:** PySpark, Hadoop, Google BigQuery, Apache Beam
- **Cloud & Deployment:** AWS (Lambda, SageMaker, API Gateway, Cloud9, CodePipeline, CodeBuild), GCP
- **Data Visualization & BI:** Tableau, Matplotlib, Seaborn
- **DevOps & MLOps:** CI/CD Pipelines, Model Optimization, Hyperparameter Tuning
- **Tools & APIs:** Yahoo Finance API, LangChain, AstraDB, Gemini API, Flask

Professional Experience

- Data Scientist, LPL Financial** 02/2024 – Present | Remote, USA
- Partnered with financial analysts and investment teams to define model objectives, gathering requirements for stock price prediction and integrating real-time market data using Python, SQL, and Yahoo Finance API.
 - Designed and deployed a deep learning-based LSTM model in TensorFlow/Keras for time-series forecasting, leveraging historical stock data to improve prediction accuracy by 12%, enabling data-driven investment strategies.
 - Developed a Reinforcement Learning (DQN)-based trading strategy, optimizing reward functions, learning rates, and policy gradients to enhance decision-making, outperforming traditional models by 15% CAGR.
 - Conducted extensive backtesting on real market data, validating stock predictions against benchmark indices, reducing prediction errors by 20%, and ensuring robustness through Monte Carlo simulations.
 - Automated model deployment on AWS Lambda, integrating API-based real-time market feeds and enabling dynamic retraining to adjust strategies based on changing financial trends.
 - Built interactive Power BI dashboards to visualize model predictions, performance metrics, and real-time stock trends, facilitating better investment decisions for stakeholders.
- Data Scientist, Tiger Analytics** 06/2022 – 07/2023 | Chennai, India
- Analyzed large datasets from MS SQL Server and MongoDB using SQL and Python. Provided insights into financial performance, supporting budgeting and forecasting activities with detailed analytical reports.
 - Applied machine learning algorithms (ARIMA, K-means Clustering) to forecast sales and optimize inventory management. Enhanced inventory accuracy and reduced holding costs by 15%.
 - Utilized Apache Spark/PySpark and Hadoop for processing and analyzing large-scale financial data. Implemented data pipelines using Apache Airflow and NiFi for efficient data ingestion and processing.
 - Developed and maintained interactive dashboards and reports using Tableau and Qlik Sense. Presented key financial metrics and trends to stakeholders, facilitating data-driven decision-making.
 - Deployed data analytics solutions on AWS and Azure, leveraging Docker and Kubernetes for containerization and orchestration. Implemented CI/CD pipelines with Jenkins for continuous integration and deployment.
 - Collaborated with finance and IT teams to ensure data accuracy and integration. Utilized tools like JIRA and GitHub for project management and version control.
- Associate Data Scientist, Genpact** 05/2021 – 05/2022 | Tirupati, India
- Collaborated with cross-functional teams in Agile sprints, conducting requirement gathering sessions to align the customer churn prediction model with business needs and stakeholder expectations for impactful insights.
 - Developed and fine-tuned a customer churn prediction model using Random Forest, XGBoost, and Logistic Regression, achieving 92% accuracy through advanced hyperparameter tuning and feature engineering techniques.
 - Conducted extensive data preprocessing on 50,000+ customer records, handling missing values, outlier detection, and encoding categorical variables using Pandas, NumPy, and Scikit-learn for optimal model training.
 - Evaluated models using metrics like ROC-AUC, Precision-Recall, and performed 10-fold cross-validation, improving recall by 15% through iterative testing and hyperparameter optimization using GridSearchCV.
 - Deployed the model using Amazon SageMaker and AWS Lambda, integrating real-time API endpoints via API Gateway and ensuring scalable production-ready solutions with secure deployment pipelines.
 - Automated CI/CD pipelines with AWS CodePipeline and CodeBuild, incorporating real-time model performance monitoring and automated updates for efficient production management.

Education

- Master of Science, University at Buffalo-The State University of New York** 08/2023 - 12/2024 | Buffalo, NY, USA
Data Science
- Bachelors' in Technology, Indian Institute of Technology Tirupati** 08/2018 - 05/2022 | Tirupati, India
Electrical Engineering

Projects

- **AI-Powered Chatbot for Trading Insights Oct 2024**
Built an end-to-end AI chatbot using Flask, LangChain, AstraDB, and Gemini API, integrating a Retrieval-Augmented Generation (RAG) model to deliver real-time trading insights with a 95% response accuracy, handling 500+ concurrent queries.
- **Metric Learning for Image Similarity Search Nov 2022**
Developed a self-supervised machine learning pipeline with a hybrid loss function (Triplet Loss and Categorical Cross Entropy) and optimized data pipelines, improving image similarity search precision by 15% and model accuracy by 20%.